



Comprehensive study of corrosion mechanisms in Integrated Circuit (IC) copper metal lines within TEM lamellae

Céline Borowiak^{*}, Frédéric Lorut, Jean-Gabriel Mattei

STMicroelectronics Crolles, 850 Rue Jean-Monnet, Crolles 38920, France

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ABSTRACT

Corrosion of copper circuitry features embedded in a Transmission Electron Microscope (TEM) lamella prepared from semiconductor devices can be a significant issue for process monitoring, as it can cause the observed patterns to be either masked or distorted, and also as it may modify the structure and composition of samples. Impacted TEM lamellae have been tracked for several months, with no correlation from sample preparation equipments nor human behavior. To understand corrosion mechanisms, in-depth studies were performed to evidence elements responsible for copper segregation using Energy-Dispersive X-Ray Spectroscopy (EDS) or Electron Energy Loss spectroscopy (EELS), revealing the presence of chlorine, sulfur and oxygen elements. Specific tracking of air composition of the laboratory indicated no excess of contaminant species. The use of protective coatings and structure grounding during Focused Ion Beam (FIB) preparation of the TEM lamella have been tested and revealed to be efficient to eradicate corrosion.

1. Introduction

Latest technology nodes of semiconductor ICs embed copper circuitry. To support either the manufacturing of new products or the development of new process, physical characterization is required to monitor and analyze/check the stacked layers leading to a final circuit. Among the existing techniques available in a physical characterization laboratory, Transmission Electron Microscopy enables to characterize and visualize features at the nanometer scale. This technique requires to shape the material to be observed into a thin foil, called TEM lamella. The most appropriate technique employed to prepare such specimens is to use a system combining both a FIB and a Scanning Electron Microscope (SEM) (Giannuzzi and Stevie, 1999), as it allows to target a specific feature within an IC structure, while enabling to thin the TEM lamellae down to a typical 100 nm thickness (all samples mentioned in the present study have been prepared by using this FIB method).

When dealing with TEM lamellae embedding copper circuitry layers, diffusion parasitic shapes originating from copper features have been observed. They can be a major problem for transmission electron microscopy analyses (often aiming at observing crystalline structures and nanoscale materials), as they degrade the morphological observation of IC copper patterns, but also because they may alter the structure and composition of samples.

Few studies have been reporting this phenomenon (Seidel et al., n.d.; Bender et al., 2005), with the effect being either named ‘corrosion’ or ‘degradation’ in the literature. The described phenomenon will be named hereafter corrosion.

To gauge the range of copper corrosion, the number of impacted TEM lamellae has been monitored for several months, with the aim to identify possible root causes. In addition, several studies have been conducted to investigate the mechanisms of corrosion and the contributing factors. The results of these studies have helped to better understand the causes of corrosion in copper-containing lamellae, leading to setup different methods to prevent or treat corrosion.

2. Experimental details

Copper corrosion study has been conducted in two distinct axes: on one side, tracking of the presence of copper contamination on the TEM lamellae of the analyzed structures/technology nodes to determine and analyze the frequency of corrosion occurrence, and to identify possible commonalities among impacted samples. In addition to this data analysis work, to evaluate the various parameters that may influence the corrosion process, and to assess the relevance of containment solutions, a dedicated structure - ensuring a 100 % occurrence of corrosion, has been processed. The structure is a patterned back-end copper layer, on

^{*} Corresponding author.

E-mail address: celine.borowiak@st.com (C. Borowiak).

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which a resist layer has been deposited and encapsulated with an additional copper capping. The sample is named hereafter copper corrosion sensitive structure.

All samples have been prepared using various gallium FIB/SEM systems from Thermo Fisher (Helios600, Helios450, H1200AT and Helios G5, all equipped with loading ports) using in-situ lift out method (Giannuzzi and Stevie, 1999) on TEM grids (molybdenum and copper TEM grids), to achieve a typical 100 nm thick lamella. FIB milling conditions used to prepare TEM lamellae were the following: a 30 kV acceleration voltage and current in the [22; 47]nA range was used for rough milling; then, pre-shaping was performed with a 2.5 nA current. Once lifted and welded onto a TEM grid, the current was lowered from 0.79 nA down to 80pA as the sample was getting close to the final thickness. For some trials, a low-kV showering has been used, with a 5 kV acceleration voltage and 50pA current. Once sample is prepared and unloaded from the FIB/SEM system, it is placed into an anti-ESD TEM grid box, itself placed into an anti-ESD container, stored in a cupboard at ambient air in the laboratory (the atmosphere in the laboratory is temperature and hygrometry controlled).

TEM analyses were performed on a Thermo Fisher Tecnai F20 operating at 200KV, while Scanning Transmission Electron Microscopy (STEM) experiments were performed on a Tecnai OSIRIS operating at 200 kV fitted with the spectrometer Enfina Gatan energy filter for EELS, and superX detectors for EDS mappings. Electron Loss Near Edge Spectroscopy (ELNES) lines profiles were performed at 0.3 eV/channel. STEM micrographs and EELS data were acquired and processed under Gatan Digital Micrograph environment. STEM EDS were processed using

Esprit software.

3. Results

A typical microelectronic circuitry is made up with stacked copper metal lines as presented in the cross-section TEM lamella, Fig. 1a. When corrosion occurs, the electrical circuitry layers exhibit various shapes of copper degradation. Among these, needles appear to scatter everywhere on the TEM cross section (Fig. 1b), while dendrites shapes can be observed surrounding the copper metal line (Fig. 1c); additionally, some dots or nanoparticles were identified on these impacted lamellae (Fig. 1d).

TEM lamellae exhibiting copper corrosion patterns have been tracked every week for a seven-month long period (representing a dataset of more than 1500 observed TEM lamellae). The resulting number of occurrences of impacted TEM lamellae enabled to build up the trendline of the corrosion issue presented Fig. 2. Obviously, only TEM lamellae embedding copper structures were affected. Among them, an average of 25 % of analyzed structures with back-end copper circuitry were corroded (reaching up to 40 % for some weeks).

No commonality related to the structure nor technology node could be made (surprisingly, even two TEM lamellae welded on the same TEM grid could present different behavior, i.e. one being impacted while the other being free from copper corrosion).

The characterization workflow has also been questioned to identify possible commonalities related to the impacted TEM lamellae; however, no evidence has been demonstrated when related to the equipment used

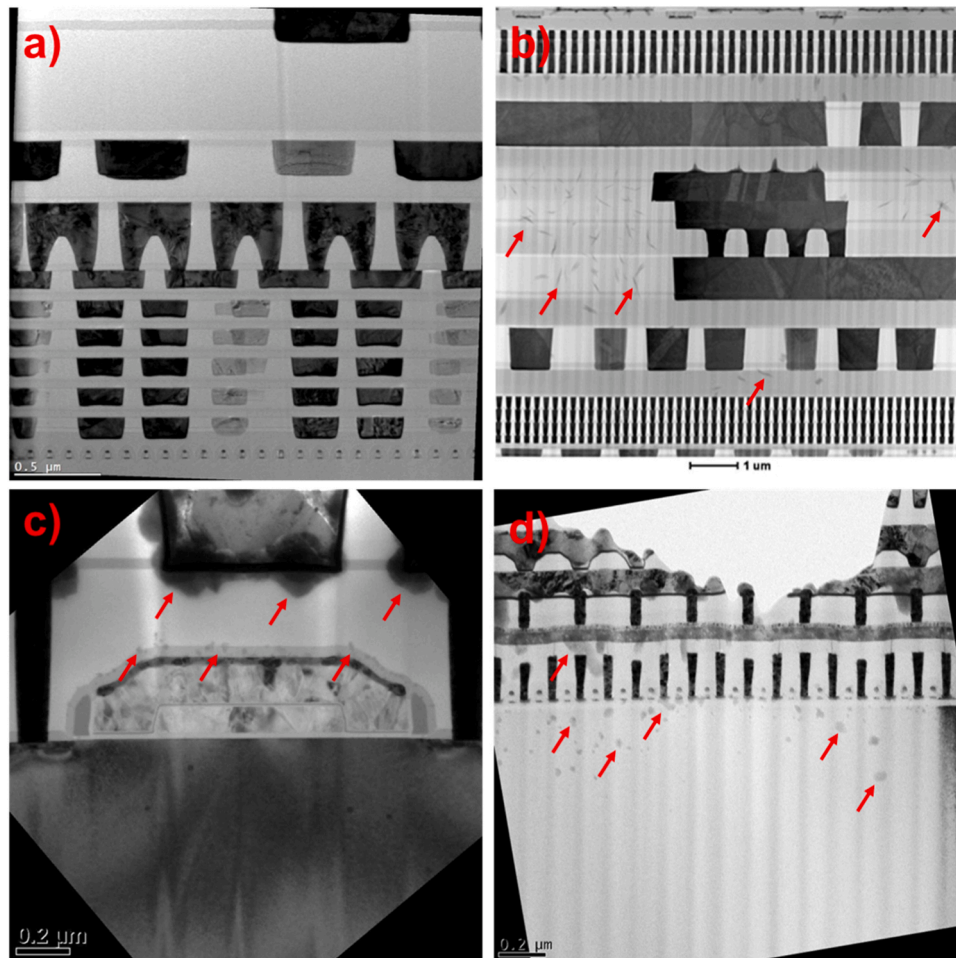


Fig. 1. a) Copper metal line free from corrosion, b) Copper metal corroded with needles scattered along the lamella, c) Corrosion along a metal line (dendrites) and d) Corrosion induced dots/particles.

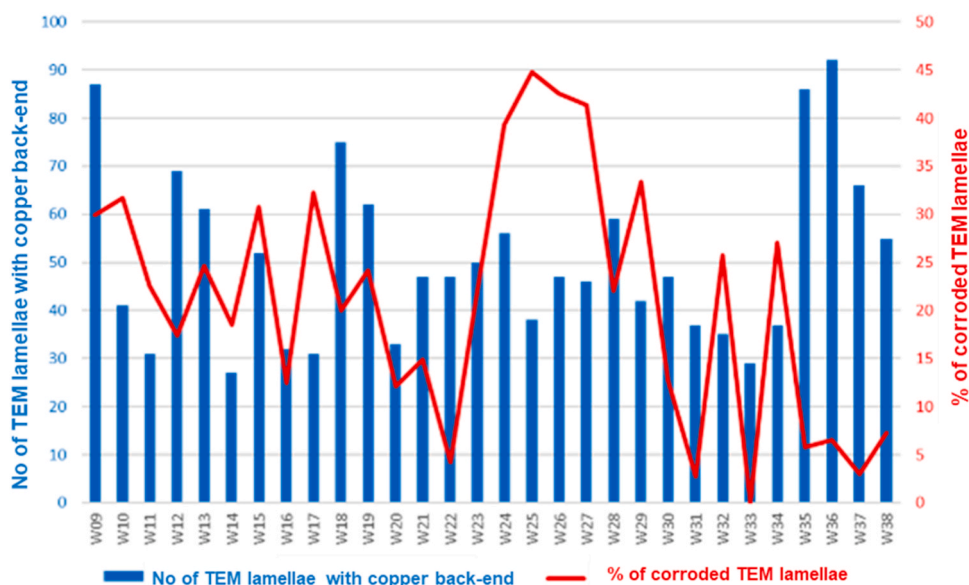


Fig. 2. Trendline of copper back-end corroded TEM lamellae, showing the number of TEM lamellae with copper back-end circuitry prepared and observed every week (blue bars), and the percentage of corroded TEM lamellae among them (red line).

for sample preparation: as corrosion is observed whatever the FIB/SEM equipment used, no evidence of a marginal FIB/SEM equipment chamber can be pointed out. In addition, copper corrosion was present whatever the FIB conditions used to prepare the TEM lamellae (various gallium source voltage and current used for the final thinning step, respectively 30 kV down to 5 kV, and 200pA down to 50pA), sample preparation process could not be highlighted as the root cause of the corrosion.

Copper corrosion has been reported for many analysts manipulating samples, while clean room constraints have been respected (wear of gloves and anti-Static discharge suits). Therefore, this corrosion cannot be attributed to a human behavior.

Specific attention has been made on TEM grids (use of new copper grids and molybdenum grids) and their storage (ESD boxes or membrane boxes), without any change.

No copper degradation on TEM lamella has been observed during

sample preparation in the FIB/SEM system, using SEM imaging. Corrosion has been observed by SEM imaging only for prepared samples reloaded within FIB/SEM systems after being exposed to ambient air (not very often as TEM lamellae are mainly prepared once in a row), or by TEM imaging once samples were exposed to air. The influence of timelapse between the shaping of a TEM lamella and its observation has been pointed out as worsening the copper diffusion (point later discussed Fig. 10). The waiting time could differ, as TEM observation could be achieved directly after sample preparation or sample could be stored few days before TEM analysis. Unfortunately, no ‘time’ rule could be made, defining a critical time value ensuring a corrosion issue, as copper degradation has been reported with samples with no waiting time.

For a better understanding of the origin of copper corrosion patterns, a large panel of TEM based techniques have been employed.

First, to examine the chemical nature of these corrosion patterns, scanning TEM EDS was performed to determine the species present in a

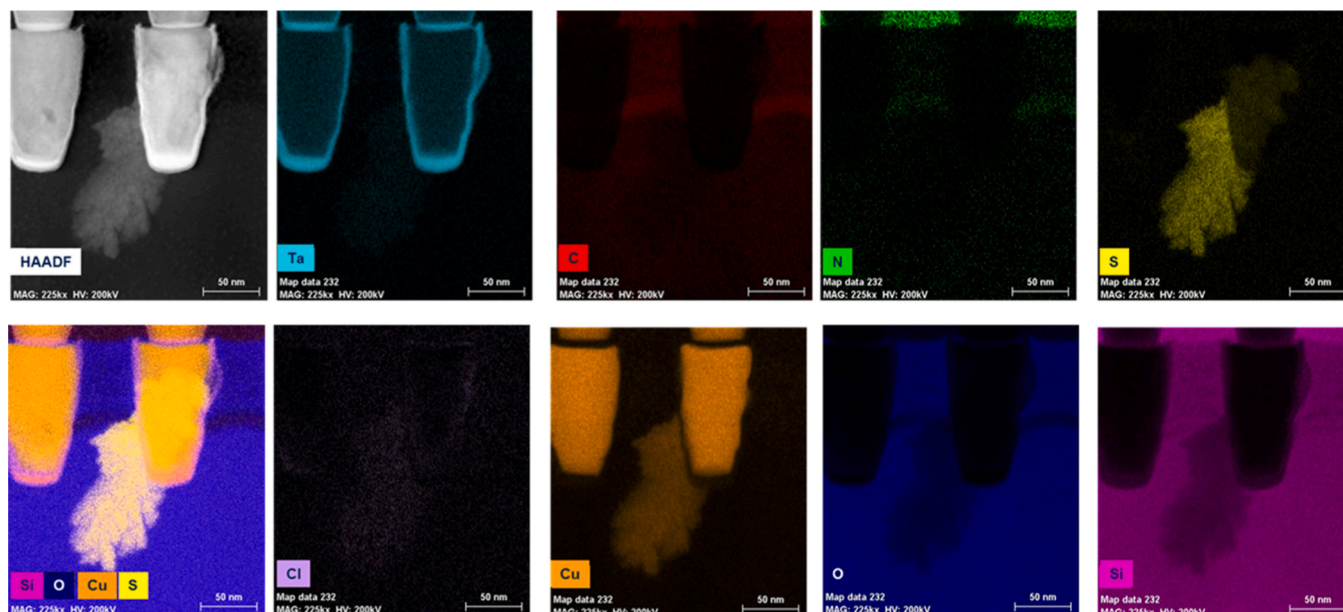


Fig. 3. STEM EDS acquisition into copper metal line corroded, with Ta, C, N, S, Cl, O and Si mappings displayed in net intensities.

corroded defect. A metal line (composed of a core of copper separated from the SiO₂ dielectric by a Tantalum barrier) with copper segregation has thus been analyzed. While copper element presence in the defect has been confirmed, a clear track of sulfur and oxygen were identified inside the segregation, as well as possible traces of chlorine (Fig. 3). Such an EDS mapping has been repeated for multiple corrosion defects; for some cases, chlorine has not been observed, while sulfur remained present.

EELS spectroscopy is a powerful technique that can provide information on the chemical composition and especially bonding states of materials (Mattei et al., 2023). It has thus been employed to describe the chemical environment of copper within a corrosion defect, on a 100 nm thick TEM lamella.

Hence, a STEM EELS line scan was performed across the defect at both oxygen and copper edges (Fig. 4). The fine structure of the oxygen ionization K edge was examined across the SiO₂/Cu interface to detect any environmental change. A line profile was performed across the metal/SiO₂ dielectric interface, and EELS spectra were extracted from three regions: 50 nm away from the interface in the SiO₂ layer (spectrum 3), 15 nm away from the interface (spectrum 2), and at the interface (spectrum 1). Fig. 3b and c illustrate respectively these three spectra and present oxygen K and copper L_{2,3} ionization edges. Spectra 3 and 2, acquired in the SiO₂ layer, exhibit a typical fine structure of oxygen bonded to Si atom, validating the SiO₂ environment. However, the oxygen edge of the spectrum acquired at the Cu/SiO₂ interface appears to be energetically shifted, indicating a different environment than SiO₂. Moreover, the oxygen fine structure is also different from the one in a SiO₂ environment; this may signify oxygen is bonded with copper element. To confirm this hypothesis, the fine structure of the copper L_{2,3} edges was investigated in 3 areas: one spectrum was acquired at the Cu/SiO₂ interface, a second spectrum was acquired inside the copper metal line and a third one 15 nm away from the metal line, where diffusion appears to occur (as shown in Fig. 3c). The spectrum extracted from the Cu metal line displays a specific fine structure of Cu metal, indicating Cu-Cu bonding. However, at the interface and in the diffusion area, spectra 1 and 2 exhibit typical white lines specific to oxidation (Hofer and Golob, 1987; Wang et al., 2016; Mattei et al., 2023; Ayoub et al., 2023). While the literature suggests the specific shape of spectrum 2 to be related to the fine structure of Cu₂O, spectrum 1 shape could be interpreted as copper being bonded to either S or a CuO composition. This result confirms observations made at the oxygen K edge, which suggests that oxygen was bonded to an element other than silicon. Therefore, this observation clearly reveals copper oxidation at the Cu/SiO₂ interface and beyond (specifically in the diffusion area) and points out a Cu₂O phase and/or a CuS phase. Nevertheless, STEM imaging provides a 2D projection of a 3D material, therefore it cannot be concluded if copper corroded segregates into the core of the lamella or along the surface. In answering this question, a cross section was performed across this TEM lamella along the copper segregation and

located apart from copper metal lines (i.e. in the dielectric layer neighboring the corroded metal lines).

Schematic of the in-plane TEM lamella preparation from the existing cross-section TEM lamella is presented Fig. 5: first an electron beam assisted deposition of carbon is performed to protect the surface to be observed. Then a platinum ion beam assisted deposition is added to protect the s

tack during the shaping of the in-plane lamella. Once thinned, the obtained lamella consists of a thin SiO₂ layer, in the middle of a stack made on both sides of a carbon layer, and a platinum layer.

As illustrated in the Fig. 6a), a STEM HAADF image displays the cross section of the previous TEM lamella, encapsulated by carbon capping. The dark region is associated to the SiO₂ layer, while the brighter region on its right (in between the SiO₂ layer and the carbon capping) corresponds to the corrosion segregation. As the observed lamella is an in-plane observation, this means that this corrosion patterns is located at the surface of the previous observed cross section TEM lamella. Also, since STEM HAADF is a Z contrast imaging technique, the bright contrast suggests a heavier element – which could be attributed to copper. A STEM EELS mapping has been carried out in this region (green rectangle on Fig. 6a); as the energy window for EELS does not allow to acquire energy losses from low energy (oxygen K -edge at 535 eV and silicon L -edge at 99 eV) to higher energy losses ranges, we preferred to opt for the energy range including copper L_{2,3}, silicon K and sulfur K ionization edges, respectively at 935, 1830 and 2470 eV. These observations and chemical mappings point out that copper corrosion process occurs on the surface of the lamellae, suggesting this phenomenon to be related to the interaction of the lamellae with the external environment.

Efforts have thus been made to identify possible key parameters leading to this corrosion phenomenon.

3.1. Occurrence

Copper corrosion process could involve different parameters such as humidity, energy (or inner electrical fields or charges) and air contaminants (S, Cl and O) (Seidel et al., n.d.; Bender et al., 2004). The influences of all these specific parameters have been investigated.

Regarding humidity and laboratory temperature, sample preparation takes place in a controlled environment in which humidity and temperature are constantly tracked and monitored (Fig. 7). The recorded levels did not correlate with the observed corrosion crisis occurrence. It can also be noticed that no specific increase of copper corrosion was triggered during heavy rain period nor during warmer seasons.

To assess the possible evolution of sulfur and chlorine levels in ambient air that could match the copper corrosion events, a monitoring of these contaminants has been set up: every week, a new blank silicon wafer has been exposed to ambient air, and then analyzed using ToF-Sims technique (using ToF-Sims 5 equipment from IONTOF), focusing

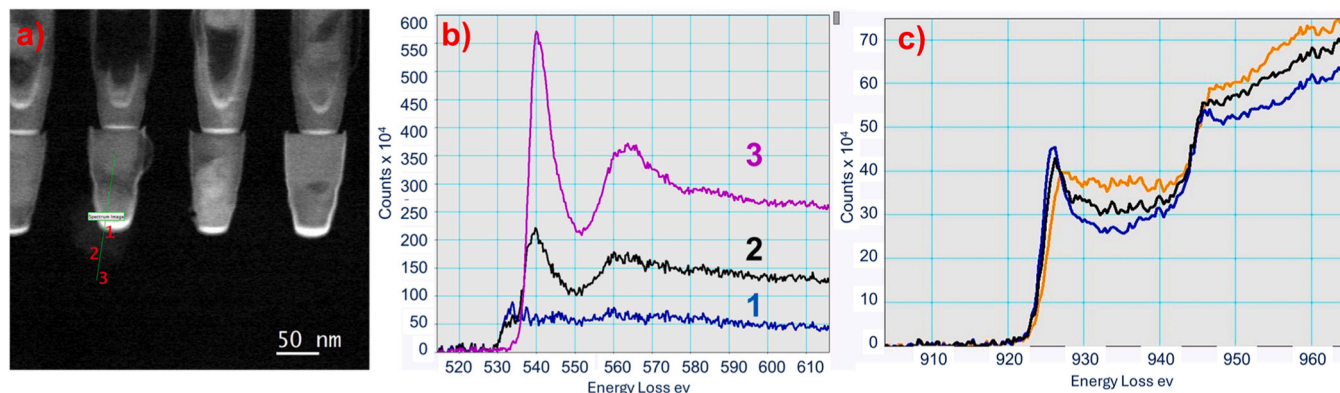


Fig. 4. STEM EELS at oxygen K and copper L_{2,3} edges.

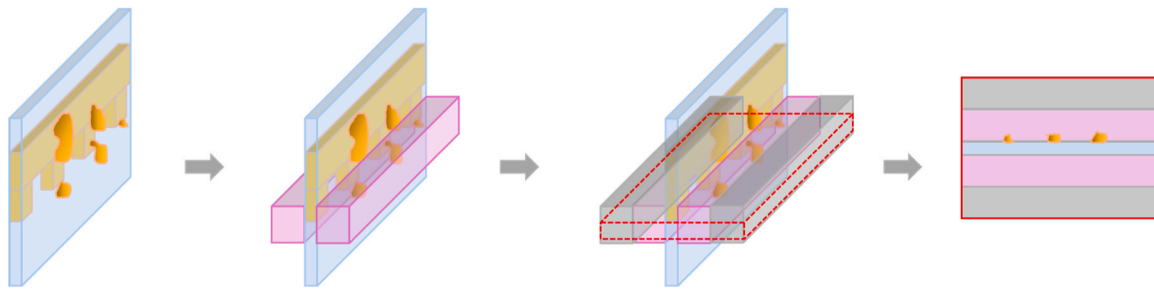


Fig. 5. Schematic of TEM lamella workflow, showing a) the initial lamella, b) the e-beam carbon deposit performed on the two sides of the lamella, c) the added platinum capping layer and the targeted position of the in-plane lamella, d) the resulting stack.

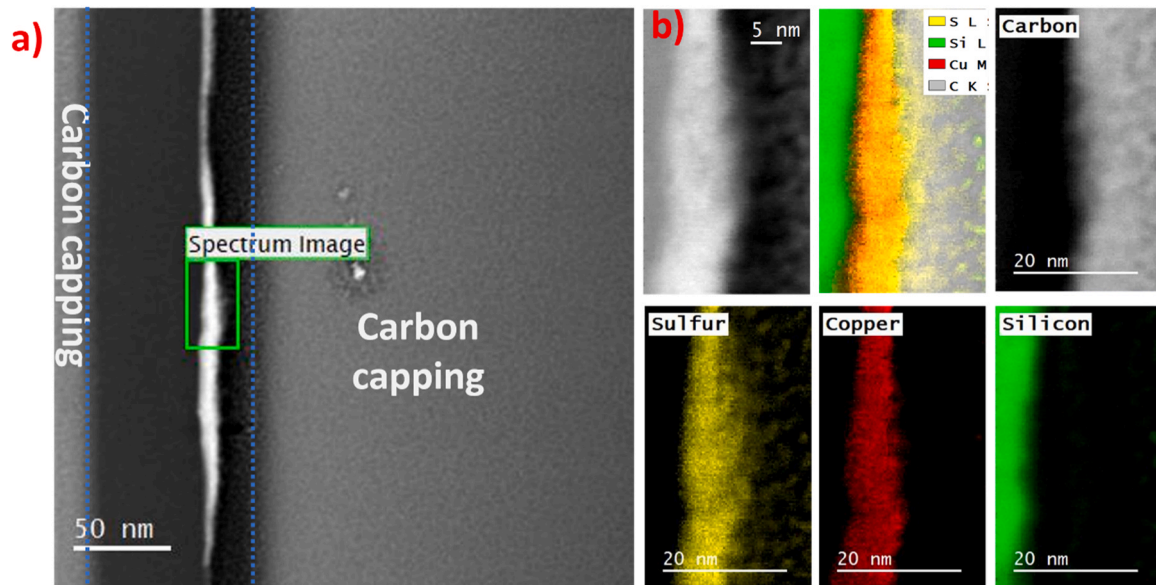


Fig. 6. a) EELS mapping of planar section b) mapping of sulfur, copper and silicon.

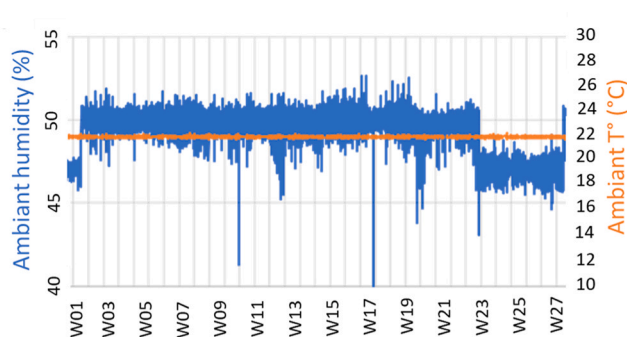


Fig. 7. Monitoring of temperature and humidity for a six-month period.

on sulfur and chlorine fragments; the operation has been repeated during few weeks, and the results are presented Fig. 8 and Fig. 9. The recorded levels did not correlate with the copper diffusion peaks observed.

The same analysis has been conducted once in another ST laboratory where no corrosion issue has been reported. The measured sulfur level was in the same range as the one in our laboratory. This pointed out that the presence of sulfur in the ambient air, even if a possible origin of corrosion, could not be considered as the sole cause of the corrosion problems.

As no obvious external parameter (humidity, temperature or

contaminants levels) could be evidenced to correlate the monitored corrosion crisis, and as there is no trivial link of the corrosion effect with technology nodes, it is likely that the corrosion process is related to the lamella itself. Sulfur/chlorine could be already present on the lamella due to previous exposure or contamination during the manufacturing process, but in undetectable quantities (not evidenced by EDS nor EELS). To initiate the corrosion process, the needed energy could be related to the presence of charges or inner electric field (either related to the device observed, or due to the FIB/SEM process).

3.2. Aging study

The worsening of the corrosion effect with increased timelapse between TEM lamella preparation and TEM imaging has already been highlighted when discussing the monitoring of impacted TEM lamellae. To assess copper diffusion process as a function of time, a TEM lamella has been prepared from the copper diffusion sensitive structure (described in experimental details section), and immediately followed by a TEM inspection; After a 8 days long storage in a TEM grid box (both membrane box and anti-ESD box have been tested), the lamella has been observed again. From these two TEM images, the following statement could be made: while clean after its preparation, a long storage affects the copper structure integrity with severe occurrences of copper diffusion observed (Fig. 10). To validate the sample sensitivity to TEM lamellae corrosion, the experience has been repeated multiple times, with the same invariable effect observed. This stacked layers sample has thus been employed to evaluate the benefit of possible containments to

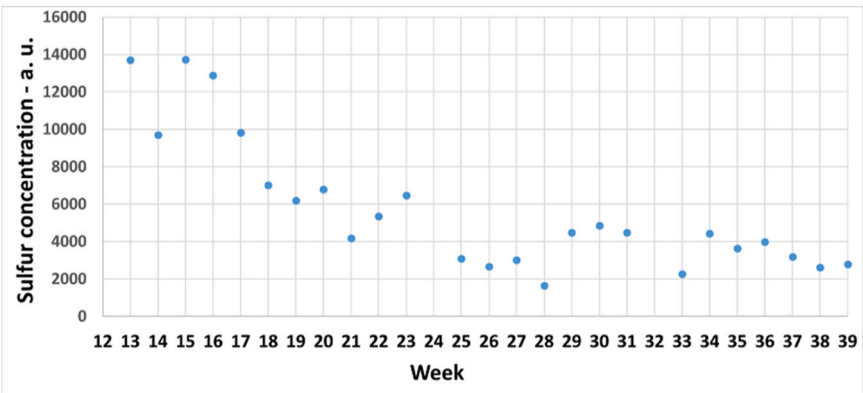


Fig. 8. Monitoring of S content of the ambient air at laboratory.

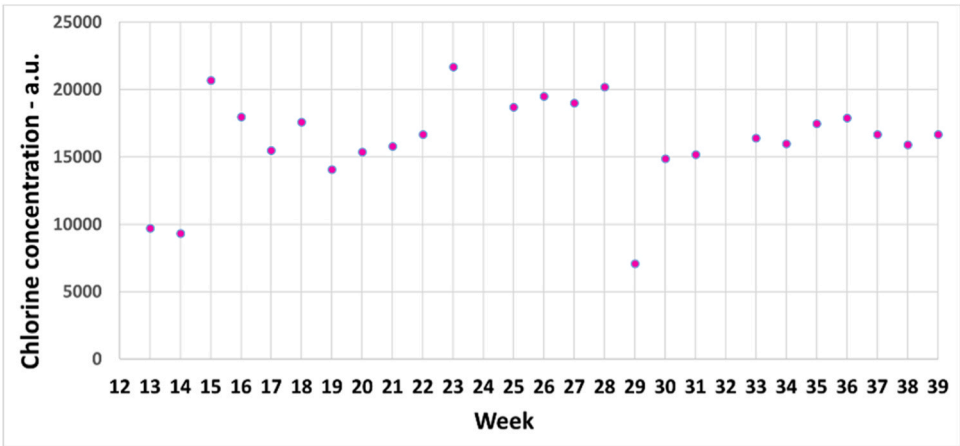


Fig. 9. Monitoring of Cl content of the ambient air at laboratory.

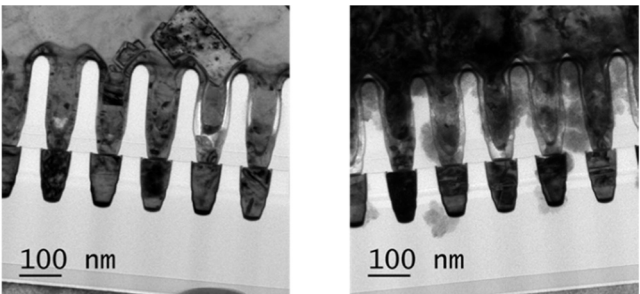


Fig. 10. copper diffusion aging - after sample preparation (left), after a 8 days storage (right).

eradicate this corrosion effect.

3.3. Possible containments

For one part, we evaluated the benefit of a capping performed on both sides of the lamellae; another strategy consisted of grounding copper structures with silicon substrate. Details of both containments and obtained results are presented here below.

3.4. Capping containment

As the corrosion process is likely to start once the lamella is exposed to ambient air, the capping has been performed inside the FIB/SEM system, as a final step of sample preparation. This protocol has already

been proposed in previous studies (Ricci et al., 2015; Seidel et al., 2016). We applied the same methodology, with following parameters: an electron beam assisted deposition has been performed to obtain a carbon capping (using naphthalene gaz precursor). A 5 kV accelerating voltage, with a typical 1 nA current has been used to expose a typical 5 μm x 2 μm surface for a minute, on both sides of the lamellae. The resulting capping was only few nanometers thick and confirmed to be transparent during TEM observation.

A lamella has been prepared from the copper diffusion sensitive sample, with multiple thinned regions: one with no capping (for reference), another one with the proposed carbon capping. The two regions have been observed one day after their FIB preparation to confirm no corrosion was observed, and again after a 9 days long storage in a

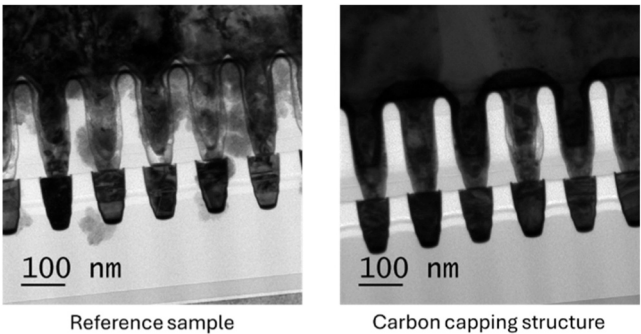


Fig. 11. Benefit assessment of Carbon capping (right) compared to standard (left) TEM lamella preparation.

conventional TEM grid box. The obtained TEM images after this storage period are presented Fig. 11; the benefit of carbon capping to prevent any copper diffusion process is clearly evidenced.

Note: another capping material transparent to electrons has also been evaluated, namely an insulator TEOS-like capping, using ethyl-silicate gas precursor. Same results as for carbon capping have been obtained (carbon was preferred as available on all FIB/SEM systems, which was not the case for TEOS-like material).

This capping result strongly highlights the influence of air as major factor leading to copper diffusion.

3.5. Grounding

As the copper corrosion effect is an electrochemical process, it is somehow related to electrical charges (or electric field) within the TEM lamellae, leading to copper diffusion. To stop this copper flow, the copper pattern embedded in the lamella has been grounded, bridging the copper layer with the silicon substrate: this has been achieved thanks to a platinum ion beam assisted deposition just before the last thinning steps. This platinum strap has been performed on one side of the lamellae and placed on the edges of the thinned part of the lamellae. The obtained lamella has been observed one day after sample preparation, and then 9 days after standard storage. No copper diffusion tracks were observed on this lamella, as shown in Fig. 12.

While proven to be effective on our test sample, this protective method could be difficult to setup when dealing with complex copper architecture, difficult to ground without being damaged or shadowed.

3.6. Other trials

Other solutions have been tested, with unsuccessful results. A plasma cleaner step within the FIB/SEM chamber has been tested, without any effect on copper corrosion effect. Also tested, a UV cure of TEM lamellae just after unloading from the FIB/SEM tool did not offer any benefit.

4. Conclusion

Corrosion of copper metal lines within TEM lamellae has been investigated by TEM imaging. Tracking number of impacted TEM lamellae over months allowed to claim no statistical correlation related to equipment nor human factor could be made but pointing out the influence of time as a worsening factor. Characterization of corrosion patterns has been conducted: chemical EDS mappings indicate the presence of sulfur (and sometimes chlorine); these results have been reinforced by EELS analysis, which reveals copper corrosion to be related to the formation of CuO compounds. In-plane characterization of impacted TEM lamellae highlights this corrosion shapes to be located on the surface of the lamellae, not in the core, suggesting the influence of external parameters. Sulfur contaminants in the air have been thus tracked for weeks, without any correlation between sulfur trend and number of corroded TEM lamellae. Assumption was made that sulfur leading to corrosion is present in the lamella itself, although in undetectable quantities, and activated thanks to external triggers. Monitoring of air laboratory with respect to humidity and temperature has been implemented, with no trend corrosion matching.

Without any proven corrosion parameter identified, pragmatic containment strategies have been tested to eradicate the corrosion phenomenon: while some trials revealed themselves inoperant (UV cure, in FIB/SEM plasma cleaning), structure grounding enabled to avoid any corrosion issue, and suggested the presence of electrical charges as another influencing factor. Capping the sides of the lamellae with a carbon coating, as a final step of sample preparation also managed to prevent corrosion. This step could be easily implemented in the sample preparation workflow and allowed to eradicate the corrosion issue within our laboratory.

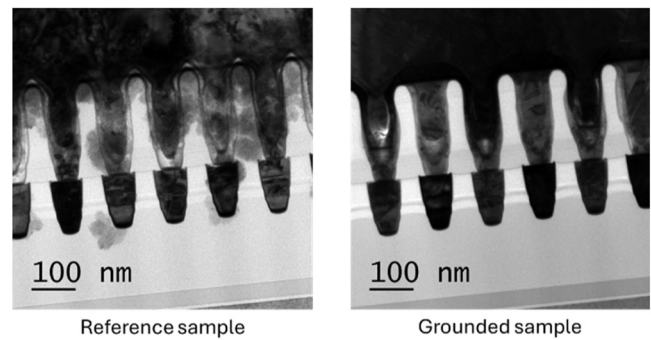


Fig. 12. TEM images of copper diffusion sensitive structure after 9 days of storage: standard preparation (left) and grounded copper layers (right).

Authors contribution statement

Céline Borowiak did the overall monitoring and lead corrosion mechanisms investigation. Frederic Lorut performed specific sample preparation and contributed to the paper writing. Jean-Gabriel Mattei performed EELS and EDX analysis and contributed to the paper writing. All authors were involved in the preparation of the manuscript. All authors have read and approved the final manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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